

Electromagnetic Force and Torque in a DC Motor

1 Force on a Current-Carrying Conductor

A conductor carrying an electric current in a magnetic field experiences an electromagnetic force.

In vector form,

$$\boxed{\mathbf{F} = I \boldsymbol{\ell} \times \mathbf{B}}$$

where

- $\mathbf{F}$ is the electromagnetic force,
- I is the electric current,
- $\boldsymbol{\ell}$ is the length vector of the conductor,
- $\mathbf{B}$ is the magnetic flux density.

The current I is a *scalar*. The direction of the current is represented by the direction of $\boldsymbol{\ell}$. The magnitude of the force is therefore

$$F = I\ell B \sin \theta,$$

where θ is the angle between $\boldsymbol{\ell}$ and $\mathbf{B}$.

If the conductor is perpendicular to the magnetic field,

$$\theta = 90^\circ,$$

and hence

$$\boxed{F = BI\ell}$$

since $\sin 90^\circ = 1$.

2 From Force to Torque

Torque is produced when a force acts at a distance from the axis of rotation.

In vector form,

$$\boxed{\mathbf{T} = \mathbf{r} \times \mathbf{F}}$$

where $\mathbf{r}$ is the position vector from the axis of rotation to the point at which the force acts. Its magnitude is

$$T = rF \sin \alpha.$$

For the usual geometry of a DC motor, the electromagnetic force acts approximately perpendicular to the radius, so

$$T = rF.$$

Using

$$F = BI\ell,$$

we obtain

$$T = rBI\ell.$$

Thus, for a motor of fixed geometry,

$$\boxed{T \propto BI}$$

because r and ℓ , together with the number and arrangement of conductors, are fixed by the construction of the motor.

3 Magnetic Flux and Magnetic Flux Density

For a uniform magnetic field perpendicular to an area A ,

$$\boxed{\phi = BA}$$

where

- ϕ is the magnetic flux,
- B is the magnetic flux density,
- A is the area through which the magnetic field passes.

For a motor of fixed geometry, A is constant. Therefore,

$$B \propto \phi.$$

Substituting this relationship into

$$T \propto BI,$$

gives

$$\boxed{T \propto \phi I}$$

and for the armature of a DC motor,

$$\boxed{T = K_T \phi I_a}$$

where

- T is the electromagnetic torque,
- K_T is the torque constant,
- ϕ is the magnetic flux per pole,
- I_a is the armature current.

Therefore, the electromagnetic torque of a DC motor is proportional to both the magnetic flux and the armature current.

4 Back EMF

When the armature rotates in a magnetic field, an electromotive force is induced in the armature conductors.

This induced voltage opposes the applied voltage and is therefore called the *back electromotive force* (back EMF).

For a DC motor,

$$E = K_E \phi N$$

where

- E is the back EMF,
- K_E is the EMF constant,
- ϕ is the magnetic flux per pole,
- N is the rotational speed.

Hence,

$$E \propto \phi N$$

For constant magnetic flux,

$$E \propto N$$

so the back EMF increases as the motor rotates faster.

5 Armature Voltage Equation

Applying Kirchhoff's voltage law to the armature circuit gives

$$V = E + I_a R_a$$

or equivalently,

$$E = V - I_a R_a$$

where

- V is the terminal voltage,
- E is the back EMF,
- I_a is the armature current,
- R_a is the armature resistance.

Thus,

$$I_a = \frac{V - E}{R_a}$$

The term $I_a R_a$ is an actual resistive voltage drop, whereas E is an induced voltage caused by the rotation of the armature.

6 Mechanical Power

For translational motion,

$$W = Fx.$$

For rotational motion, the corresponding relation is

$$\boxed{W = T\theta}$$

where θ is the angular displacement.

Dividing by time,

$$P = \frac{W}{t} = T\frac{\theta}{t}.$$

Since angular velocity is

$$\omega = \frac{\theta}{t},$$

the mechanical power is

$$\boxed{P = T\omega}$$

Thus, the correspondence between translational and rotational motion is

$$\boxed{\begin{array}{ll} W = Fx & \longleftrightarrow W = T\theta, \\ P = Fv & \longleftrightarrow P = T\omega. \end{array}}$$

7 Summary

The important relationships for a DC motor are

$$\boxed{\mathbf{F} = I\boldsymbol{\ell} \times \mathbf{B}}$$

$$\boxed{T = K_T\phi I_a}$$

$$\boxed{E = K_E\phi N}$$

$$\boxed{V = E + I_a R_a}$$

and

$$\boxed{P = T\omega.}$$

The physical chain can be summarized as

Magnetic field + armature current $\longrightarrow$ electromagnetic force $\longrightarrow$ torque $\longrightarrow$ rotation.

At the same time, rotation of the armature in the magnetic field produces the back EMF:

magnetic flux + rotation $\longrightarrow$ back EMF.